PIPES

TECHNICAL SPECIFICATION REFERENCE -STEEL PIPES

Excerpts from JIS G 3452 & ASTM A-53 Specifications

Classification	Specification	Grade	Chemical Composition					Tensile Test				Bend Test *1		Flattening
			C max.	Mn max.	Si max.	P max.	S max.	Tensile Strength	Yield Strength	Elongation		Bending Angle	Bending Radius	Test H
			%	%	%	%	%	N/mm²	N/mm²	%				
Carbon Steel Pipes for Ordinary Piping	JIS G 3452	SGP	--	--	--	0.04	0.04	290 min.	--	* Test Piece No. 11 & 12 * Test Piece No. 5	30 min. 25 min.	90°	6D	$^{2}/_{3}$ D
ASTM Standard	ASTM A53	A	0.25	0.95	--	0.05	0.06	330 min.	205 min.	As specified in A-53 specification		90°	12D	$^{1}/_{3}$ D
Welded Steel Pipes	ASTM A53	B	0.30	1.20	--	0.05	0.06	415 min.	240 min.					

Notes:

* - When the tensile test is carried out on No. 12 or No. 5 test piece for the pipe under 8 mm in wall thickness, the minimum value of elongation shall be obtained by subtracting 1.5 % from the thickness values of elongation given in Table above for each 1 mm decrease in wall thickness, and rounding off to an integer in accordance with JIS Z 8401.

* - The values of elongation given in Table above shall not applied to the pipe whose nominal size js 32 mm or smaller.

*1 - Bend test in table above only applied to pipes of nominal size 2" (50 mm) or smaller.

H - Distance between the plates

D - Outside diameter of the pipe